Practice workbook · Define a balanced diet and explain why needs differ between people

Meaning of a balanced diet and its components, and differences in energy or nutrients with sex, age, pregnancy, menstruation or infancy.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-BALANCED-DIET-AND-DIFFERING-NEEDS

1. Question 2 · part a · 2 marks

November 2024 Paper 2

Worked-example retry · Attempt part a only.

2 (a) Describe what is meant by a balanced diet.

(2)

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

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1. Question 2 · part a · 2 marks · Working and answer

Working and answer

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2. Question 2 · part a · 2 marks

June 2019 Paper 1R

Further practice · Attempt part a only.

2 Humans need a balanced diet.

(a) Vitamins are needed in a balanced diet.

State four other components of a balanced diet.

(2)

1 .....

2 .....

3 .....

4 .....

(b) Explain why vitamin C is needed in a balanced diet.

(2)

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2. Question 2 · part a · 2 marks · Working and answer

Working and answer

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3. Question 4 · part b · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

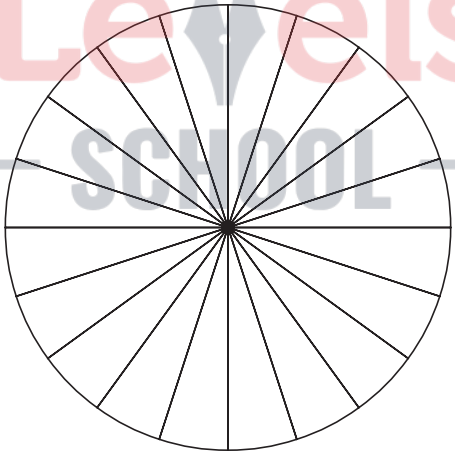
4 The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

Component	Percentage (%) composition of diet	
	Area A	Area B
cereals	75.0	15.0
fruit and vegetables	15.0	37.5
eggs, fish and meat	2.5	27.5
milk, cheese and butter	7.5	20.0

(a) Complete the pie chart to show the diet of the girl in Area B.

(4)



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(b) Explain what is meant by a balanced diet.

(2)

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3. Question 4 · part b · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b only.

(c) Calculate the ratio of the percentage composition of cereals in the diet of the girl in Area A compared with the diet of the girl in Area B.

Give your answer in the form n : 1

(2)

ratio = ...

(d) Discuss the effects that the percentage composition given for eggs, fish and meat in the diet may have on the growth of the two girls.

(5)

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(Total for Question 4 = 13 marks)

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3. Question 4 · part b · 2 marks · Working and answer

Working and answer

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4. Question 3 · part a.i · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.i only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
MALES					
15 to 18 years old	12 600	59	0.9	50	12
25 to 50 years old	12 200	63	0.9	60	10
FEMALES					
15 to 18 years old	9200	46	0.7	50	15
25 to 50 years old	9200	50	0.7	60	15

(i) Describe what is meant by a balanced diet.

(2)

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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4. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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5. Question 2 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

2 (a) Describe what is meant by a balanced diet.

(2)

(b) A label on a pack of four slices of roast beef gives information about how much of each dietary component is present in the meat.

It also gives the percentage of the recommended daily amount for each dietary component.

Table 1 and table 2 give these values.

Energy in kJ	Fat in g	Sugars in g	Salt in g
470	2.4	less than 0.4	0.6

Table 1

Percentage of recommended daily amount (%)			
energy	**fat**	**sugars**	**salt**
4.0	4.0	less than 1.0	12.0

Table 2

(i) Name another dietary component that could be listed on the pack of meat.

(1)

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5. Question 2 · part b.i · 1 mark

November 2024 Paper 2

Further practice · Attempt part b.i only.

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(ii) Using data from the tables, calculate the mass of fat, in g, that is needed to achieve the recommended daily amount.

(3)

mass = ... g

(iii) Calculate the number of slices of meat a person would need to eat to supply a recommended daily energy intake of 11 800 kJ.

(2)

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number of slices = ...

P 7 5 9 5 9 A 0 5 2 4

Turn over ▶

5. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

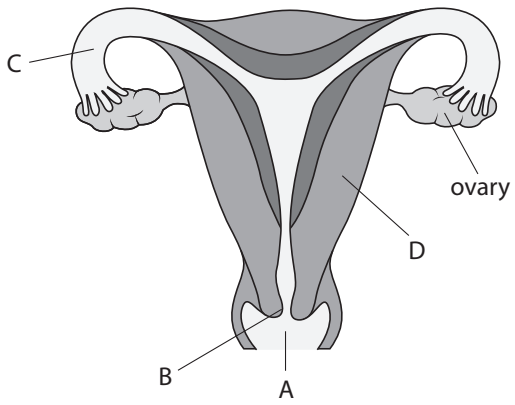
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6. Question 2 · part b · 1 mark

January 2020 Paper 1

Further practice · Attempt part b only.

2 (a) The diagram shows some structures of the female reproductive system.



(i) Where are sperm deposited during sexual intercourse?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

(ii) Which part of the female reproductive system contracts during birth to push the fetus downwards?

(1)

- ☒ **A**
- ☒ **B**
- ☒ **C**
- ☒ **D**

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6. Question 2 · part b · 1 mark

January 2020 Paper 1

Further practice · Attempt part b only.

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(iii) The ovaries produce female sex hormones.

The box lists words associated with the female reproductive system.

FSH	wall	fertilisation	oestrogen	ovulation
lining	ADH	follicles	menstruation	

Use words from the box to complete the passage about the ovaries.

(3)

The ovaries produce a hormone called ..... .

This hormone repairs the uterus ..... following menstruation. Once a month, one egg is released from the ovaries. This is known as ..... .

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(b) State why a pregnant female should have more calcium in her diet than a female who is not pregnant.

(1)

.....

.....

(c) Some women breastfeed their babies.

Give one advantage of breastfeeding babies rather than feeding them bottled milk.

(1)

.....

.....

6. Question 2 · part b · 1 mark · Working and answer

Working and answer

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7. Question 5 · part a.ii · 4 marks

November 2020 Paper 1R

Further practice · Attempt part a.ii only.

5 (a) The type and amount of different nutrients needed for a balanced diet varies depending on several factors.

(i) Pregnant women are often advised by their doctor to take extra iron in their diet.

Which two foods are the best sources of iron?

(1)

- ☒ **A** fruit and green vegetables
- ☒ **B** dairy products and red meat
- ☒ **C** green vegetables and red meat
- ☒ **D** fruit and dairy products

(ii) Explain why a woman should have more iron in her diet when she becomes pregnant.

(4)

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7. Question 5 · part a.ii · 4 marks

November 2020 Paper 1R

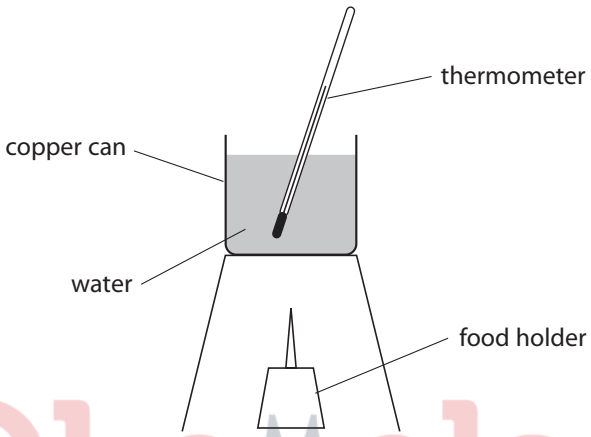
Further practice · Attempt part a.ii only.

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(b) The diagram shows some apparatus that can be used to measure the amount of energy contained in different foods.



(i) Explain how the apparatus can be used to obtain data about the energy content of different foods.

(4)

(ii) Explain how this apparatus can be changed to improve the accuracy of the data collected for each food.

(3)

(Total for Question 5 = 12 marks)

7. Question 5 · part a.ii · 4 marks · Working and answer

Working and answer

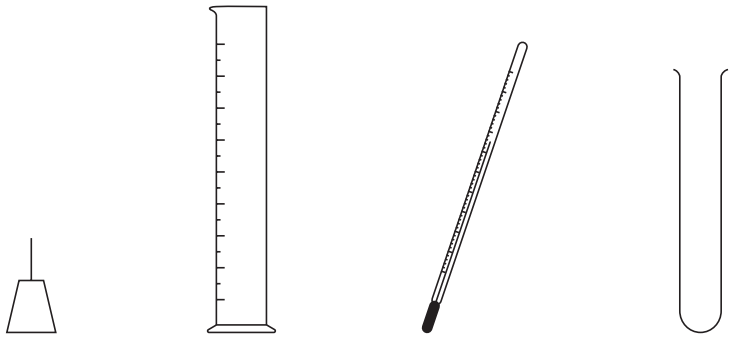
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8. Question 3 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

3 (a) The diagram shows some equipment that can be used to investigate the energy content of different foods.



mounted needle | measuring cylinder | thermometer | test tube

(i) Describe how the equipment can be used to investigate the energy content of different foods.

(5)

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8. Question 3 · part b.ii · 1 mark

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Further practice · Attempt part b.ii only.

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(ii) Name two variables that need to be controlled in the investigation.

(2)

1 .....

.....

2 .....

.....

(iii) Give two safety precautions needed for this investigation.

(2)

1 .....

.....

2 .....

.....

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8. Question 3 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

(b) Different people have different energy requirements.

The table shows the daily energy requirements in kilocalories (kcal) of males and females of different ages.

Daily Energy Requirement in kcal

Age of adults in years	Males	Females
19–24	2772	2175
25–34	2749	2175
35–44	2629	2103
45–54	2581	2103
55–64	2581	2079
65–74	2342	1912
75 and over	2294	1840

(Source: adapted from https://propranolols.blogspot.com/2020/02/daily-nutritional-requirements-for_9.html)

(i) Describe the patterns in the energy requirements for males and for females.

(2)

(ii) Which of these gives a reason for the difference between the energy required by 25-year-old males and the energy required by 25-year-old females?

(1)

- ☒ **A** females use more muscle mass than males
- ☒ **B** females eat less than males
- ☒ **C** males eat less than females
- ☒ **D** males use more muscle mass than females

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8. Question 3 · part b.ii · 1 mark

June 2022 Paper 1

Further practice · Attempt part b.ii only.

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(iii) Calculate the percentage difference in the energy requirement of males compared with females in the age range of 35–44 years.

(3)

percentage difference = ... %

(Total for Question 3 = 15 marks)

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Working and answer

9. Question 3 · part a.ii · 3 marks

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
MALES					
15 to 18 years old	12 600	59	0.9	50	12
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FEMALES					
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(i) Describe what is meant by a balanced diet.

(2)

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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9. Question 3 · part a.ii · 3 marks

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Further practice · Attempt part a.ii only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

(2)

(iv) State a nutrient, shown in the table, that can help to reduce night blindness.

(1)

(v) Red meat and some green vegetables are good sources of iron.

100 g of red meat contains 2.5 mg of iron.

100 g of broccoli contains 0.70 mg of iron.

A male in the age range 15 to 18 years eats 250 g of red meat and 200 g of broccoli in one meal.

Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

(4)

extra mass = ... mg

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9. Question 3 · part a.ii · 3 marks · Working and answer

Working and answer

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10. Question 3 · part a.iii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.iii only.

3 (a) The table gives the recommended daily dietary intake for different groups of males and females.

Group	Energy in kJ	Protein in g	Vitamin A in mg	Vitamin C in mg	Iron in mg
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15 to 18 years old	12 600	59	0.9	50	12
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(i) Describe what is meant by a balanced diet.

(2)

(ii) Explain the difference between the amount of energy needed by males compared with females in the age range 25 to 50 years.

Use data from the table to help your answer.

(3)

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10. Question 3 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

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(iii) Explain why females in the age range 25 to 50 years need more iron in their diet than males in the same age range.

(2)

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(1)

.....

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Calculate the extra mass, in mg, of iron the male needs to eat to meet his recommended daily dietary intake of iron.

(4)

extra mass = ... mg

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10. Question 3 · part a.iii · 2 marks · Working and answer

Working and answer

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